

Backpropagation, Part 2: How It Runs in Code

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Gradient descent ([Note 57](#)); Sigmoid function ([Note 74](#)); Derivatives of one variable ([Note 600](#)); Partial derivatives and gradients ([Note 601](#)); Forward propagation ([Note 1010](#)); Loss functions in deep learning ([Note 1014](#)).
- **Leads to:** Vanishing gradient ([Note 1018](#)); Improving a neural network ([Note 1021](#)).

1. Overview

Key point: We turn the algorithm into code twice: for regression (linear nodes, squared error) and for classification (sigmoid nodes, binary cross-entropy). Our code and Keras, started from the same weights, produce the same losses to six digits.

The [backpropagation what Note](#) built the algorithm and its 9 derivatives for a 2-2-1 regression network. This Note runs it:

1. **Regression:** the same network and data, trained for several epochs with our own NumPy code, then with Keras.
2. **Classification:** the same architecture with sigmoid activations and binary cross-entropy. The derivatives change, so we derive them, then train again.

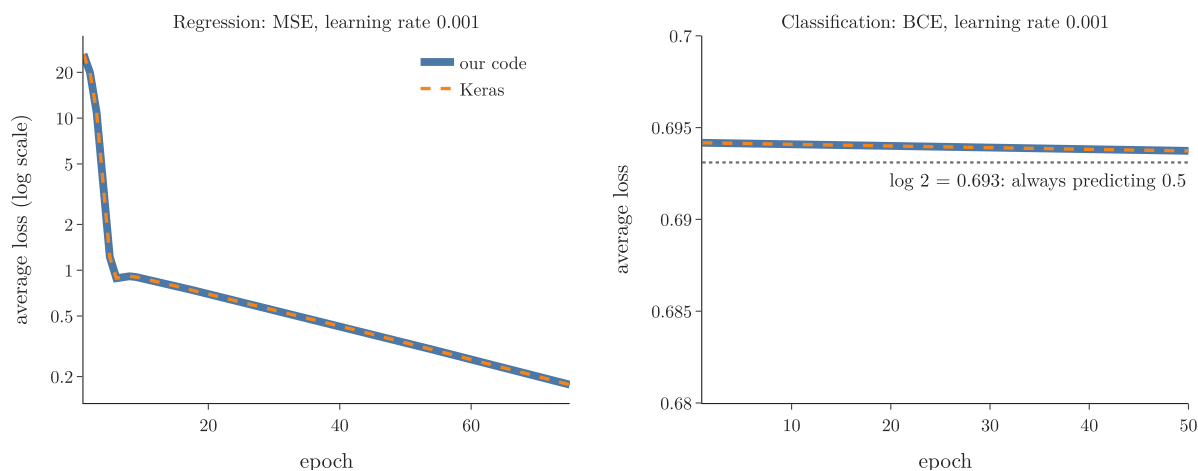


Figure 1: Average loss per epoch for our code and for Keras, started from the same weights

Figure 1 shows the result: in both problems the two lines lie on top of each other. Keras runs the same backpropagation we write by hand.

2. Prerequisites

- The [backpropagation what Note](#): the algorithm, the network and the regression derivatives.
- The [sigmoid derivative Note](#) and the [logistic regression gradient Note](#), for the classification derivatives.

3. The regression algorithm, in loops

Key point: An outer loop over epochs, an inner loop over the 4 students; inside: forward, loss, update all 9 parameters. At the end of each epoch, average the 4 losses.

The data and network are those of the [backpropagation what Note](#): four students (CGPA, profile score, package), a 2-2-1 network with linear nodes, all weights 0.1 and biases 0. The algorithm:

1. Initialise the parameters.
2. For each of 5 epochs:
 - for each student, in order:
 - a. predict $\hat{y}$ with forward propagation;
 - b. compute the loss $(y - \hat{y})^2$;
 - c. update the 9 parameters with $W \leftarrow W - \eta \partial L / \partial W$, $\eta = 0.001$.
 - print the average of the 4 losses: the loss of the epoch.

Students are taken in order here. In practice the rows are usually picked in a random order each epoch, which the [gradient descent in neural networks Note](#) returns to.

4. The regression code

Key point: Four small functions: initialise, forward, update, train. The update function is the 9 derivative formulas, one line per group.

4.1 Initialising and predicting

Key point: One weight matrix and one bias vector per layer; forward propagation keeps the hidden outputs, because the update needs them.

Python: Initialisation and forward propagation.

```
import numpy as np

def initialize_parameters(layer_dims):
    # weights 0.1, biases 0; W[l] has shape (nodes in l-1, nodes in l)
    return {l: [np.full((layer_dims[l-1], layer_dims[l]), 0.1),
                 np.zeros(layer_dims[l])]
            for l in range(1, len(layer_dims))}
```

```
def forward_regression(x, params):
    a1 = params[1][0].T @ x + params[1][1]      # 011, 012
    y_hat = params[2][0].T @ a1 + params[2][1]   # 021
    return y_hat[0], a1
```

`initialize_parameters([2, 2, 1])` creates a 2×2 matrix W^1 , a bias vector of length 2, a 2×1 matrix W^2 and one bias: the 9 parameters. Each layer is the matrix product of the [forward propagation Note](#), without the sigmoid.

For student 1 (8, 8) the function returns $\hat{y} = 0.32$ and the hidden outputs $a^1 = (1.6, 1.6)$. We keep a^1 because the derivatives of the output weights are $-2(y - \hat{y}) O_{11}$ and $-2(y - \hat{y}) O_{12}$.

4.2 Updating the parameters

Key point: Compute $g = -2(y - \hat{y})$ once; every update is g times the parameter's own factors.

Python: The 9 updates.

```
def update_regression(params, x, y, y_hat, a1, lr):
    g = -2 * (y - y_hat)                # dL/dy_hat
    W2 = params[2][0][:, 0].copy()      # keep the old W2
    params[2][0][:, 0] -= lr * g * a1    # W2_11, W2_21
    params[2][1][0] -= lr * g           # b21
    params[1][0] -= lr * np.outer(x, g * W2) # four W1
    params[1][1] -= lr * g * W2         # b11, b12
```

`-=` changes the arrays in place, so `params` holds the new values afterwards. The copy of `W2` matters: the hidden-layer derivatives use the weights from before this update.

The minus signs in the formulas and the minus of the update rule cancel: every update here adds $\eta \cdot 2(y - \hat{y})$ times the parameter's factors.

4.3 Training

Key point: Five epochs bring the average loss from 26.35 down to 1.22.

Python: The training loop.

```
def train_regression(X, Y, epochs=5, lr=0.001):
    params, history = initialize_parameters([2, 2, 1]), []
    for epoch in range(epochs):
        losses = []
        for x, y in zip(X, Y):          # one student at a time
            y_hat, a1 = forward_regression(x, params)
            losses.append((y - y_hat) ** 2)
            update_regression(params, x, y, y_hat, a1, lr)
        history.append(np.mean(losses))  # loss of the epoch
```

```
return params, history
```

The first epoch is exactly the one worked by hand in the [backpropagation what Note](#): losses 13.54, 21.29, 30.43 and 40.12, average 26.35. Then:

Epoch	1	2	3	4	5
Average loss	26.35	19.86	10.88	3.78	1.22

After 5 epochs the weights have moved from 0.1 to $W_{11}^1 = W_{12}^1 = 0.272$, $W_{21}^1 = W_{22}^1 = 0.393$, $W_{11}^2 = W_{21}^2 = 0.469$, with biases 0.028 (hidden) and 0.122 (output). The predictions are 5.14, 5.25, 5.36 and 5.84 LPA for packages of 4, 5, 6 and 7. They are in the right range and rising in the right order, but still too close together. After 75 epochs the average loss is 0.177 (Figure 1, left).

5. The same training in Keras

Key point: Same architecture, same starting weights, plain SGD with the same learning rate, one row per update, no shuffling: Keras gives the same losses and weights.

Python: The regression network in Keras, with our starting weights.

```
import keras

model = keras.Sequential([
    keras.Input(shape=(2,)),
    keras.layers.Dense(2, activation="linear"),
    keras.layers.Dense(1, activation="linear"),
])
model.set_weights([np.full((2, 2), 0.1), np.zeros(2),
                    np.full((2, 1), 0.1), np.zeros(1)])
model.compile(optimizer=keras.optimizers.SGD(learning_rate=0.001),
              loss="mse")
model.fit(X, Y, epochs=5, batch_size=1, shuffle=False)
model.summary() shows 9 trainable parameters. Keras normally starts from random
weights; get_weights returns the arrays in the order kernel, bias, kernel, bias, and
set_weights overwrites them. batch_size=1 updates after every row, as our loop does;
shuffle=False keeps the rows in order.
```

Keras prints the epoch losses 26.346, 19.859, 10.882, 3.777 and 1.224, and ends with exactly our weights (0.272, 0.393, 0.469, ...). The two curves in Figure 1 (left) coincide for all 75 epochs.

Extra: Every setting must match to get identical numbers. By default `fit` shuffles the rows every epoch and uses batches of 32 rows, so with default settings Keras follows a different path and ends at a slightly different loss. With an optimizer such as Adam

instead of plain SGD, the steps themselves change.

6. The classification problem

Key point: Same architecture, but every node uses a sigmoid and the loss is binary cross-entropy. Two extra steps appear in forward propagation: each weighted sum goes through the sigmoid.

Now the output is whether the student is placed (1) or not (0):

Student	CGPA	Profile score	Placed
1	8	8	1
2	7	9	1
3	6	10	0
4	5	5	0

Two things change:

- **Activations:** every node computes its weighted sum z and then outputs $\sigma(z)$, as in the [forward propagation Note](#).
- **Loss:** [binary cross-entropy](#), $L = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$, the right loss for two classes with a sigmoid output (see the [loss functions Note](#)).

The algorithm (loops, forward, loss, update) stays the same. Only the derivatives change.

For student 1, with all weights 0.1 and biases 0:

$$z_{11} = 0.1 \times 8 + 0.1 \times 8 = 1.6, \quad O_{11} = O_{12} = \sigma(1.6) = 0.832$$

$$z_f = 0.1 \times 0.832 + 0.1 \times 0.832 = 0.166, \quad \hat{y} = \sigma(0.166) = 0.5415, \quad L = -\log 0.5415 = 0.613$$

7. The classification derivatives

Key point: The sigmoid and the log cancel: $\partial L / \partial z_f = -(y - \hat{y})$. The output layer then looks like regression with -1 in place of -2 ; the hidden layer gains a factor $O(1 - O)$ from its own sigmoid.

7.1 The output layer

Key point: Three links, $L \rightarrow \hat{y} \rightarrow z_f \rightarrow W_{11}^2$; the first two multiply to $-(y - \hat{y})$.

gradient travels from the loss back to the weight; each arrow multiplies by its local derivative

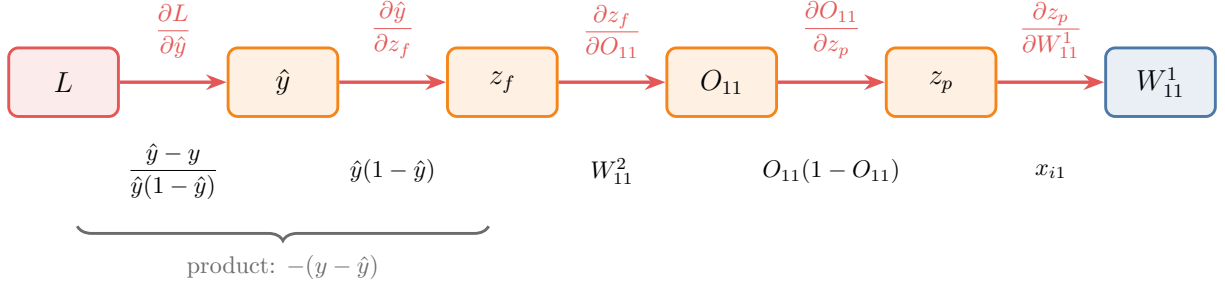


Figure 2: The chain from the loss to a first-layer weight in the classification network: each arrow multiplies by its local derivative

Call $z_f = W_{11}^2 O_{11} + W_{21}^2 O_{12} + b_{21}$ the weighted sum of the output node, so $\hat{y} = \sigma(z_f)$. A change in W_{11}^2 changes z_f , then $\hat{y}$, then L :

$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_f} \cdot \frac{\partial z_f}{\partial W_{11}^2}$$

The first two factors (Figure 2, left):

- Differentiating the loss:

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

- The sigmoid's derivative (see the [sigmoid derivative Note](#)): $\partial \hat{y} / \partial z_f = \hat{y}(1 - \hat{y})$.

Multiplied, the $\hat{y}(1 - \hat{y})$ cancels, the same simplification as in section 4.3 of the [logistic regression gradient Note](#):

$$\frac{\partial L}{\partial z_f} = \hat{y} - y = -(y - \hat{y})$$

The last factor is the same as in regression: $\partial z_f / \partial W_{11}^2 = O_{11}$, $\partial z_f / \partial W_{21}^2 = O_{12}$, $\partial z_f / \partial b_{21} = 1$. So:

$$\frac{\partial L}{\partial W_{11}^2} = -(y - \hat{y}) O_{11}, \quad \frac{\partial L}{\partial W_{21}^2} = -(y - \hat{y}) O_{12}, \quad \frac{\partial L}{\partial b_{21}} = -(y - \hat{y})$$

These are the regression formulas with -1 in place of -2 .

7.2 The hidden layer

Key point: Two more links: $z_f \rightarrow O_{11}$ gives W_{11}^2 ; $O_{11} \rightarrow z_p$ gives the sigmoid derivative $O_{11}(1 - O_{11})$.

For W_{11}^1 the chain is five links long (Figure 2). With $z_p = W_{11}^1 x_{i1} + W_{21}^1 x_{i2} + b_{11}$ the weighted sum of hidden node 1, so $O_{11} = \sigma(z_p)$:

$$\frac{\partial L}{\partial W_{11}^1} = \underbrace{\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z_f}}_{-(y-\hat{y})} \cdot \underbrace{\frac{\partial z_f}{\partial O_{11}}}_{W_{11}^2} \cdot \underbrace{\frac{\partial O_{11}}{\partial z_p}}_{O_{11}(1-O_{11})} \cdot \underbrace{\frac{\partial z_p}{\partial W_{11}^1}}_{x_{i1}}$$

The other five follow the same pattern. Only the last factor changes within a node (x_{i1} , x_{i2} or 1), and the second node uses W_{21}^2 and O_{12} :

Parameter	$\partial L / \partial (\text{parameter})$
$W_{11}^1, W_{21}^1, b_{11}$	$-(y - \hat{y}) W_{11}^2 O_{11}(1 - O_{11}) \times x_{i1}, x_{i2}, 1$
$W_{12}^1, W_{22}^1, b_{12}$	$-(y - \hat{y}) W_{21}^2 O_{12}(1 - O_{12}) \times x_{i1}, x_{i2}, 1$

Compared with regression, each hidden derivative has one extra factor, $O(1 - O)$: the derivative of the hidden node's own sigmoid.

7.3 Student 1 in numbers

Key point: $-(y - \hat{y}) = -0.4585$; output weights get -0.381 , hidden weights -0.0513 .

1. **In words:** compute $-(y - \hat{y})$ once, then multiply by each parameter's factors.
2. **Formula:** the formulas of Sections 7.1 and 7.2.
3. **Example:** $y = 1$, $\hat{y} = 0.5415$, $O_{11} = 0.832$:

$$-(y - \hat{y}) = -(1 - 0.5415) = -0.4585$$

$$\frac{\partial L}{\partial W_{11}^2} = -0.4585 \times 0.832 = -0.3815, \quad \frac{\partial L}{\partial b_{21}} = -0.4585$$

$$\frac{\partial L}{\partial W_{11}^1} = -0.4585 \times 0.1 \times 0.832 \times 0.168 \times 8 = -0.0513, \quad \frac{\partial L}{\partial b_{11}} = -0.0064$$

The hidden-layer gradients are about 7 times smaller than the output-layer ones: the factors $W_{11}^2 = 0.1$ and $O_{11}(1 - O_{11}) = 0.14$ shrink them. Each sigmoid between a weight and the loss multiplies its gradient by at most 0.25, an effect the [vanishing gradient Note](#) studies.

Python: The classification gradients.

```
def sigmoid(z):
    return 1 / (1 + np.exp(-z))
```

```
def gradients_classification(params, x, y, y_hat, a1):
    d_out = -(y - y_hat) # dL/dz_f
    W2 = params[2][0][:, 0]
    d_hidden = d_out * W2 * a1 * (1 - a1) # dL/dz_p, per node
    return {"W2": d_out * a1, "b2": d_out,
            "W1": np.outer(x, d_hidden), "b1": d_hidden}
```

`d_hidden` holds, for each hidden node, the derivative of the loss with respect to that node's weighted sum. Every weight entering the node multiplies it by its own input. In the Notebook, `tf.GradientTape` returns the same 9 numbers.

8. Training the classifier

Key point: The loss starts at 0.694 and is still 0.694 after 50 epochs. Keras, from the same start, does exactly the same: the code is right, and this network is stuck.

Training with the same loops and $\eta = 0.001$:

Epoch	1	2	50
Our code	0.6942	0.6942	0.6937
Keras	0.6942	0.6942	0.6937

The loss barely moves, and every student gets a probability of about 0.54. Is the code wrong? The same network in Keras (sigmoid activations, `loss="binary_crossentropy"`, same weights and settings) gives the same numbers to four digits (Figure 1, right). So backpropagation is computed correctly; the network is simply not learning.

The loss sits next to $\log 2 = 0.693$, the binary cross-entropy of predicting 0.5 for everyone: the network has learned nothing beyond “two students are placed, two are not”. Two causes add up:

- **A tiny learning rate** with tiny hidden gradients (Section 7.3): the weights hardly move.
- **Identical hidden nodes:** every weight starts at 0.1, so both hidden nodes stay the same, and the network acts as if it had one hidden node.

Extra: With $\eta = 0.1$ and 2,000 epochs the loss falls to 0.49: students 1 and 2 get 0.66 and student 3 gets 0.01, but student 4 still gets 0.66 instead of a low value. The two hidden nodes are still equal, so the network still acts as if it had a single hidden node. With $\eta = 0.5$ the steps are too big: the loss goes up in 36 of the 2,000 epochs, reaches its lowest value, 0.705, at epoch 194, and ends higher, at 0.76 (Notebook). Starting weights, learning rate and the sigmoid's small slope are exactly the problems that the Notes on improving a network address.

9. Summary

	Regression	Classification
Activations	linear everywhere	sigmoid everywhere
Loss	$(y - \hat{y})^2$	$-y \log \hat{y} - (1 - y) \log(1 - \hat{y})$
$\partial L / \partial z$ at the output	$-2(y - \hat{y})$	$-(y - \hat{y})$
Output-layer gradient	$-2(y - \hat{y}) O_{1j}$	$-(y - \hat{y}) O_{1j}$
Hidden-layer gradient	$-2(y - \hat{y}) W_{j1}^2 x_{ik}$	$-(y - \hat{y}) W_{j1}^2 O_{1j} (1 - O_{1j}) x_{ik}$
Training here	26.35 to 1.22 in 5 epochs	stuck at 0.694

- The code is the algorithm: initialise, then epochs of forward, loss, update, row by row.
- Our NumPy code and Keras (`SGD`, `batch_size=1`, `shuffle=False`, same starting weights) give identical losses and weights.
- For a sigmoid output with binary cross-entropy, $\partial L / \partial z = \hat{y} - y$.
- Each sigmoid on the way back adds a factor $O(1 - O)$, which shrinks the gradient.
- Correct backpropagation can still fail to learn: starting weights, learning rate and data size matter.

10. Key terms

Term	Meaning
<code>set_weights / get_weights</code>	Keras methods to write and read a model's weights and biases, as a list of arrays
<code>batch_size=1</code>	Keras setting that updates the weights after every single row
<code>shuffle=False</code>	Keras setting that keeps the rows in their original order in every epoch
<code>keras.optimizers.SGD</code>	Plain gradient descent in Keras, with a fixed learning rate
$\partial L / \partial z$	The derivative of the loss with respect to a node's weighted sum; every weight entering the node multiplies it by its own input